

Mar 18 01:22

Spyder (Python 3.11)

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...me/yusuf/Desktop/AI_ML_Progs/Mobile_price_prediction_project

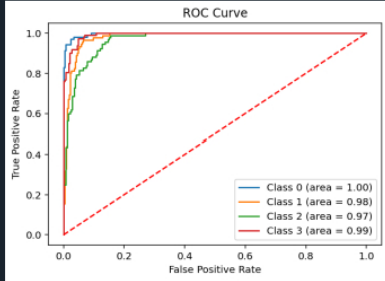
/home/yusuf/Desktop/AI_ML_Progs/diabetes_prediction_project/diabetes_project.py

diabetes_project.py

```
73 plt.ylabel("Actual")
74 plt.show()
75
76 importances = model.feature_importances_
77 features = X.columns
78
79 plt.figure(figsize=(8,5))
80 sns.barplot(x=importances, y=features)
81 plt.title("Feature Importance")
82 plt.show()
83
84 y_prob = model.predict_proba(X_test)[: ,1]
85
86 fpr, tpr, thresholds = roc_curve(y_test, y_prob)
87 roc_auc = auc(fpr, tpr)
88
89 plt.figure()
90 plt.plot(fpr, tpr, label="ROC curve (area = %0.2f)" % roc_auc)
91 plt.plot([0,1],[0,1], 'r--')
92 plt.xlabel("False Positive Rate")
93 plt.ylabel("True Positive Rate")
94 plt.title("ROC Curve")
95 plt.legend(loc="lower right")
96 plt.show()
97
98 print("\nEnter patient details to predict diabetes risk:")
99
100 preg = float(input("Pregnancies: "))
101 glucose = float(input("Glucose Level: "))
102 bp = float(input("Blood Pressure: "))
103 skin = float(input("Skin Thickness: "))
104 insulin = float(input("Insulin Level: "))
105 bmi = float(input("BMI: "))
106 dpf = float(input("Diabetes Pedigree Function: "))
107 age = float(input("Age: "))
108
109 new_data = pd.DataFrame([[preg,glucose,bp,skin,insulin,bmi,dpf,age]],
110                          columns=X.columns)
111
112 new_data_scaled = scaler.transform(new_data)
113
114 prediction = model.predict(new_data_scaled)
115
116 if prediction[0] == 1:
117     print("\nResult: HIGH RISK of Diabetes")
118 else:
119     print("\nResult: LOW RISK of Diabetes")
```

44 %

ROC Curve



Class 0 (area = 1.00)
Class 1 (area = 0.98)
Class 2 (area = 0.97)
Class 3 (area = 0.99)

Help Variable Explorer Debugger Profiler Plots Files Code Analysis

Console 1/A

0.0 0.2 0.4 0.6 0.8 1.0

False Positive Rate

Class 3 (area = 0.99)

Enter mobile features to predict price range:
Battery Power: 4500
RAM: 8192
Pixel Height: 1440
Pixel Width: 3040
Mobile Weight: 165
Talk Time: 25

Predicted Price Range: 3
Very High Cost Phone

In [9]:

IPython Console History

Inline Conda: ai_classes (Python 3.11.14) LSP: Python Line 119, Col 44 UTF-8 LF RW Mem 77%